
CMS Physics Analysis Summary

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Measurement of the WW production cross section in pp collisions at $\sqrt{s} = 8$ TeV

The CMS Collaboration

Abstract

This document reports on a measurement of the W^+W^- production cross section in pp collisions at $\sqrt{s} = 8$ TeV. The data were collected at the LHC with the CMS detector, and correspond to an integrated luminosity of 3.54 fb^{-1} . The W^+W^- candidates are selected in events with two charged leptons (electrons or muons) and large missing transverse energy. The W^+W^- cross section is measured to be 69.9 ± 2.8 (stat.) ± 5.6 (syst.) ± 3.1 (lumi.) pb.

1 Introduction

Studies of W^+W^- production test the description of electroweak and strong interactions in the standard model (SM). Next-to-leading order (NLO) calculations [1] of W^+W^- production in pp collisions at $\sqrt{s} = 8$ TeV predict a cross section of

$$\sigma^{\text{NLO}}(\text{gg} \rightarrow W^+W^- + \text{qq} \rightarrow W^+W^-) = 57.3 \left({}^{+2.4}_{-1.6} \right) \text{ pb.}$$

The dominant W^+W^- production mechanisms are the s-channel and t-channel $q\bar{q}$ annihilation diagrams. The gluon-gluon fusion contributes $\sim 3\%$. The $WW\gamma$ and WWZ triple-gauge boson couplings (TGCs) in the s-channel are sensitive to possible new physics processes at a higher mass scale. The presence of anomalous TGCs would change the W^+W^- production rate or kinematic distributions from the SM predictions. In addition, W^+W^- production is an important background source for new particle searches, e.g. Higgs boson searches [2]. In fact, the selection applied in this paper closely follows the selection applied in the Higgs boson search in the event category with zero counted jets. The Higgs boson preselection for masses above 140 GeV and the selection applied in this paper differ only in the trailing lepton minimum transverse momentum. This requirement is tightened from 10 GeV to 20 GeV to further suppress the background from $W + \text{jets}$ events, where the trailing lepton is a misidentified jet. With this selection a SM Higgs boson with a mass of 110 (130) GeV would enhance the W^+W^- signal by about 0.3% (7%).

This paper reports a measurement of the W^+W^- production cross section in pp collisions at $\sqrt{s} = 8$ TeV, performed on a dataset corresponding to an integrated luminosity of 3.54 fb^{-1} , collected during 2012.

2 CMS Detector and Simulations

While the CMS detector is described in detail elsewhere [3], the key components for this analysis are summarized here. A superconducting solenoid occupies the central region of the CMS detector, providing an axial magnetic field of 3.8 Tesla parallel to the beam direction. The silicon pixel and strip tracker, the crystal electromagnetic calorimeter and the brass/scintillator hadron calorimeter are located within the solenoid. A quartz-fiber Cherenkov calorimeter extends the coverage to $|\eta| < 5.0$, where pseudorapidity is defined as $\eta = -\ln[\tan(\theta/2)]$, and θ is the polar angle of the trajectory of the particle with respect to the beam direction. Muons are measured in gas detectors embedded in the iron return yoke outside the solenoid. The first level of the CMS trigger system, composed of custom hardware processors, is designed to select the most interesting events in less than $3 \mu\text{s}$ using information from the calorimeters and muon detectors. The High Level Trigger processor farm further decreases the event rate to a few hundred Hz, before data storage.

This measurement is based on W^+W^- candidate events in which both bosons decay leptonically. The experimental signature consists of two isolated, high transverse momentum (p_T), oppositely-charged leptons (electrons or muons) and large missing transverse energy (E_T^{miss} , defined as the modulus of the negative vector sum of the transverse momenta of all reconstructed particles, charged or neutral, in the event) due to the undetected neutrinos. Several SM processes lead to backgrounds to the W^+W^- sample. These include $W + \text{jets}$, $W\gamma$, and QCD multi-jet events where at least one of the jets is misidentified as a lepton, top production ($t\bar{t}$ and tW), Drell-Yan $Z/\gamma^* \rightarrow \ell^+\ell^-$, and diboson production (WZ , and ZZ).

A number of Monte Carlo event generators are used to simulate the signal and backgrounds.

The $q\bar{q} \rightarrow W^+W^-$ signal, $W + \text{jets}$, WZ , $W\gamma^{(*)}$, Drell-Yan and $t\bar{t}$ processes are generated using the MADGRAPH [4] event generator. The $gg \rightarrow W^+W^-$ signal component, which is expected to contribute 3% of the total W^+W^- production rate [1], is generated using GG2WW [5]. The POWHEG program [6] provided event samples for the tW process, and the remaining processes are generated using PYTHIA [7]. For leading-order generators, the default set of parton distribution functions (PDF) used to produce these samples is CTEQ6L [8], while CT10 [9] is used for next-to-leading order generators. NLO calculations are used for background cross sections. For all processes, the detector response is simulated using a detailed description of the CMS detector, based on the GEANT4 package [10].

The simulated samples are reweighted according to the distribution of number of pp interactions per bunch crossing (pile-up) as measured in the data.

3 Event Preselection

Three final states are considered: e^+e^- , $\mu^+\mu^-$ and $e^\pm\mu^\mp$. $W \rightarrow \ell\nu$ ($\ell = e, \mu$) decays are the main signal components, while $W \rightarrow \tau\nu_\tau$ events with leptonic τ decays are included, although the analysis is not optimised for this final state. The events are selected by trigger conditions requiring the presence of one or two high p_T electrons or muons.

Two oppositely-charged lepton candidates are required, both with $p_T > 20$ GeV. Muon candidates [11] are identified using a selection close to the one described in Ref. [12], while electron candidates are selected using a multivariate approach, that exploits correlations among the selection variables described in [13] to improve identification performance. The lepton candidates are required to be compatible with the primary vertex of the event, which is chosen as the vertex with highest $\sum p_T^2$ of its associated tracks. This criterion provides the correct assignment for the primary vertex in more than 99% of both signal and background events for the pile-up distribution observed in the data.

Charged leptons from W boson decays are usually isolated from other activity in the event. For each lepton candidate, a cone is constructed around the track direction at the event vertex. The scalar sum of the transverse energy of each reconstructed particle [14] compatible with the chosen primary vertex and contained within the cone is calculated excluding the contribution from the lepton candidate itself. Electron candidates are rejected if this sum exceeds 15% of the candidate p_T . Cones of several widths are constructed around the track direction at the event vertex to account for the differences in the differential energy deposit between a muon and a jet faking a lepton. This information is combined using a multivariate approach. For both electrons and muons a correction is applied to account for the contribution to the energy in the isolation cone due to the several pp interactions per bunch crossing. A median energy density (ρ) is determined event by event and the pile-up contribution to the transverse energy is estimated as the product of ρ and an effective isolation cone area. This contribution is subtracted [15] from the transverse energy in the isolation cone.

Jets are reconstructed from calorimeter and tracker information using the particle-flow technique [14, 16]. The anti- k_T clustering algorithm [17] with distance parameter $R = 0.5$ is used, as implemented in the FASTJET package [18, 19]. The jet energy is corrected for pile-up in a manner similar to the correction of the energy inside a lepton isolation cone. Jet energy corrections are also applied as a function of the jet E_T and η [20].

The properties of the hard jets are modified by particles from pile-up interactions; a combinatoric background from low- p_T jets from pile-up interactions that get clustered into high- p_T jets

may also arise. A multivariate selection is applied to separate jets coming from the primary interaction from those reconstructed from energy deposits associated with pile-up. The discrimination is based on the differences in the jet shapes, in the relative multiplicity of charged and neutral components and in the different fraction of transverse momentum carried by the hardest components. Tracks associated with a jet are required to be compatible with the primary vertex.

To reduce the background from top quark decays, events with one or more jet surviving the jet selection criteria and with corrected $E_T > 30$ GeV and $|\eta| < 4.7$ are rejected. To further suppress the top quark background, a top tagging technique based on soft-muon and b-jet tagging [21, 22] is applied. The first method vetoes events containing muons from b-quarks. The second method uses b-jet tagging applied to jets with $15 < E_T < 30$ GeV: this technique is based on tracks with large impact parameter within jets. The combined rejection efficiency for top events is about 50%.

A *projected* E_T^{miss} is defined as the component of E_T^{miss} transverse to the closest lepton if it is closer than $\pi/2$ in azimuthal angle, and the full E_T^{miss} otherwise. This observable more efficiently rejects $Z/\gamma^* \rightarrow \tau^+\tau^-$ background events, where the E_T^{miss} is preferably aligned with the leptons, as well as $Z/\gamma^* \rightarrow \ell^+\ell^-$ events with mismeasured E_T^{miss} associated with poorly reconstructed leptons or jets. Since the *projected* E_T^{miss} resolution is deteriorated by pile-up, the minimum of two different observables is used: the first includes all particle candidates in the event [14], while the second uses only the charged particle candidates associated to the primary vertex. This exploits the correlation between the two observables in events with significant real E_T^{miss} , as in the signal, and the lack of correlation otherwise, as in Drell-Yan events. In order to reduce the Drell-Yan contamination, the *projected* E_T^{miss} is required to be above 45 GeV in the e^+e^- and $\mu^+\mu^-$ final states. For the $e^\pm\mu^\mp$ final state, which has lower contamination from $Z/\gamma^* \rightarrow \ell^+\ell^-$ decays, the threshold is reduced to 20 GeV. These requirements remove more than 99% of the Drell-Yan background. To reduce the contamination from $Z/\gamma^* \rightarrow \ell^+\ell^-$ events in which the Z boson recoils against a jet, the angle in the transverse plane between the dilepton system and the most energetic jet with $E_T > 15$ GeV is required to be smaller than 165 degrees. This selection is applied only in the e^+e^- and $\mu^+\mu^-$ final states when the leading jet has $E_T > 15$ GeV. To further reduce the Drell-Yan background in the e^+e^- and $\mu^+\mu^-$ final states, events with a dilepton mass within ± 15 GeV of the Z mass are rejected. Events with dilepton masses below 12 GeV are also rejected to suppress contributions from low-mass resonances. The same requirement is also applied in the $e^\pm\mu^\mp$ final state. Finally, the transverse momentum of the dilepton system, $p_T^{\ell\ell}$, is required to be above 45 GeV to reduce both Drell-Yan and events containing jets misidentified as leptons.

To reduce the background from other diboson processes, such as WZ and ZZ production, any event that has an additional third lepton passing the identification and isolation requirements is rejected. $W\gamma$ production, in which the photon converts, is suppressed by rejecting electrons consistent with a photon conversion.

4 Estimation of Backgrounds

The W + jets and QCD multijet backgrounds are estimated from a sample of events in which one lepton passes the nominal requirements while the other passes a looser set of requirements but which fails the nominal ones. The looser requirements are defined by looser criteria on impact parameter and isolation. This sample, enriched in W + jets events, is extrapolated to the signal region using the efficiencies for such loosely-identified fake leptons to pass the tight selection. These efficiencies are measured in data using multi-jet events and are parametrized

according to the kinematics of the lepton candidate. The systematic uncertainty is about 36%, coming mainly from the estimation of these efficiencies.

The normalization of the top background that survives the top veto is estimated from data by counting top tagged events, N_{tagged} . The background from mis-tagged non top events, $N_{\text{tagged}}^{\text{bkg}}$, is subtracted, and the corresponding event top tagging efficiency is applied. The event top tagging efficiency, $\epsilon_{\text{top tagged}}$, is measured in a data control sample dominated by $t\bar{t}$ and tW events, selected by requiring one b-tagged jet with $E_T > 30$ GeV. The background from mis-tagged non top events is subtracted from both the numerator and denominator. The top background, $N_{\text{not tagged}}$, is then given by

$$N_{\text{not tagged}} = (N_{\text{tagged}} - N_{\text{tagged}}^{\text{bkg}}) \times (1 - \epsilon_{\text{top tagged}}) / \epsilon_{\text{top tagged}},$$

The systematic uncertainty on $N_{\text{not tagged}}$ is about 18%, which is dominated by the statistical uncertainty in the control sample and from systematic uncertainties related to the measurement of $\epsilon_{\text{top tagged}}$.

We estimate the Drell-Yan contribution to the e^+e^- and $\mu^+\mu^-$ final states outside of the Z mass window $N_{\text{out}}^{\ell\ell, \text{exp}}$, by normalising the simulation to the observed number of events inside the Z mass window, $N_{\text{in}}^{\ell\ell}$, after subtracting non Drell-Yan backgrounds. The backgrounds from $t\bar{t}$, W^+W^- , and WZ and ZZ events where both leptons do not come from the same Z boson, $N_{\text{in}}^{\text{non-Z}}$, are subtracted by counting the number of $e^\pm\mu^\mp$ events in the Z mass window, taking into account combinatorics and the relative detection efficiencies for electrons and muons. The WZ and ZZ background, $N_{\text{in}}^{\text{ZV}}$, where both leptons come from the same Z boson, is subtracted by using simulation. The residual Drell-Yan background in data outside the Z mass window is thus expressed as

$$N_{\text{out}}^{\ell\ell, \text{exp}} = R_{\text{out/in}}^{\ell\ell} \left(N_{\text{in}}^{\ell\ell} - N_{\text{in}}^{\text{non-Z}} - N_{\text{in}}^{\text{ZV}} \right), \text{ with } R_{\text{out/in}}^{\ell\ell} = N_{\text{out}}^{\ell\ell, \text{MC}} / N_{\text{in}}^{\ell\ell, \text{MC}}.$$

The systematic uncertainty on the final estimate is derived from the dependence of $R_{\text{out/in}}^{\ell\ell}$ on the value of the E_T^{miss} cut, by taking the difference between the $R_{\text{out/in}}^{\ell\ell}$ value in the signal region and in a region where the E_T^{miss} cut has been relaxed. The value of the systematic uncertainty is 24%.

Finally, a control sample with three reconstructed leptons is used to measure the cross section of the $W\gamma^*$ process. We only use $W\gamma^* \rightarrow \ell\nu\mu^+\mu^-$ events for this measurement because the $\ell\nu e^+e^-$ final state is difficult to separate from other backgrounds. Therefore the pair of oppositely charged leptons with the lowest invariant mass must be a pair of muons. This measurement is used to normalise the simulated $W\gamma^*$ background contribution from asymmetric gamma decays, in which one lepton escapes detection [23].

Other backgrounds, such as the WZ and ZZ diboson production, are estimated from simulation. The $W + \text{jets}$ and $W\gamma/W\gamma^*$ background estimate is cross-checked in data using the events passing all the selection requirements, except that the two leptons must have the same charge; this sample is dominated by $W + \text{jets}$ and $W\gamma/W\gamma^*$ events, after subtracting the expected WZ background, where one lepton from the Z decay is lost. The $Z/\gamma^* \rightarrow \tau^+\tau^-$ contamination is also cross-checked using $Z/\gamma^* \rightarrow e^+e^-$ and $Z/\gamma^* \rightarrow \mu^+\mu^-$ events selected in data, where the leptons are replaced with simulated τ decays, showing good agreement with expectation from simulation.

The estimated event yields from all processes after applying the event selection are summarized in Table 1, where the WW contributions are calculated assuming the SM cross section.

The distributions of the key analysis variables are shown in Figure 1, where the WW contribution is normalized according to the measured cross section.

Table 1: Observed and expected event yields in 3.54 fb^{-1} . The prediction for the W^+W^- process assumes the SM cross section value.

sample	yield $\pm$ stat. $\pm$ syst.
$gg \rightarrow WW$	$43.3 \pm 1.0 \pm 13.4$
$qq \rightarrow WW$	$640.3 \pm 4.9 \pm 47.4$
$t\bar{t} + tW$	$131.6 \pm 12.7 \pm 19.5$
$W + \text{jets}$	$60.0 \pm 4.3 \pm 21.6$
$WZ + ZZ$	$27.4 \pm 0.5 \pm 2.9$
Z/γ^*	$42.5 \pm 6.0 \pm 9.9$
$W\gamma + W\gamma^*$	$13.6 \pm 2.4 \pm 4.3$
total background	$275.2 \pm 14.9 \pm 31.2$
signal + background	$958.8 \pm 15.7 \pm 58.3$
data	1111 ± 33

5 Efficiencies and Systematic Uncertainties

The signal efficiency is estimated using simulation, considering both the $q\bar{q} \rightarrow W^+W^-$ and $gg \rightarrow W^+W^-$ processes. Correction factors for the lepton reconstruction and identification efficiency are determined using $Z/\gamma^* \rightarrow \ell^+\ell^-$ events [24] recorded with dedicated unbiased triggers. These factors depend on the lepton p_T and $|\eta|$ and are typically around 0.96 for electrons and 0.98 for muons. The trigger efficiency for signal events is measured to be approximately 98% for both $q\bar{q} \rightarrow W^+W^-$ and $gg \rightarrow W^+W^-$ final states.

The uncertainties on lepton momentum scale and resolution, E_T^{miss} modeling and jet energy scale are applied to the reconstructed objects in simulated events by smearing and scaling the relevant observables and propagating the effects to the kinematic variables used in the analysis. The uncertainty assigned to the pile-up corresponds to shifting the mean number of pile-up events per beam crossing up and down by one interaction, and amounts to 2.3%.

The uncertainty on the acceptance due to variations in the parton distribution functions and the value of α_s was found to be 2.3% by following the PDF4LHC prescription [25, 26]. The effect of higher-order corrections was found to be 1.5% by varying the QCD renormalisation (μ_R) and factorisation (μ_F) scales using the MCFM program [1].

The W^+W^- jet veto efficiencies in data are estimated from simulation, and multiplied by a data-to-simulation scale factor derived from $Z/\gamma^* \rightarrow \ell^+\ell^-$ events in the Z peak: $\epsilon_{W^+W^-}^{\text{data}} = \epsilon_{W^+W^-}^{\text{MC}} \times \epsilon_Z^{\text{data}} / \epsilon_Z^{\text{MC}}$. The uncertainty is thus factorized into the uncertainty on the Z efficiency in data and the uncertainty on the ratio of the W^+W^- efficiency to the Z efficiency in simulation ($\epsilon_{W^+W^-}^{\text{MC}} / \epsilon_Z^{\text{MC}}$). The former, which is statistically dominated, is 0.3%. Theoretical uncertainties due to higher order corrections contribute most to the WW/Z efficiency ratio uncertainty, which is estimated to be 4.6% for W^+W^- production. The data-to-simulation correction factor is found to be close to one.

The uncertainties on the $W + \text{jets}$ and top background predictions are 36% and 15%, respectively, as described in Section 4.

The theoretical uncertainties on the WZ and ZZ cross sections are calculated by varying the QCD renormalisation (μ_R) and factorisation (μ_F) scales using the MCFM program [1]. The effect

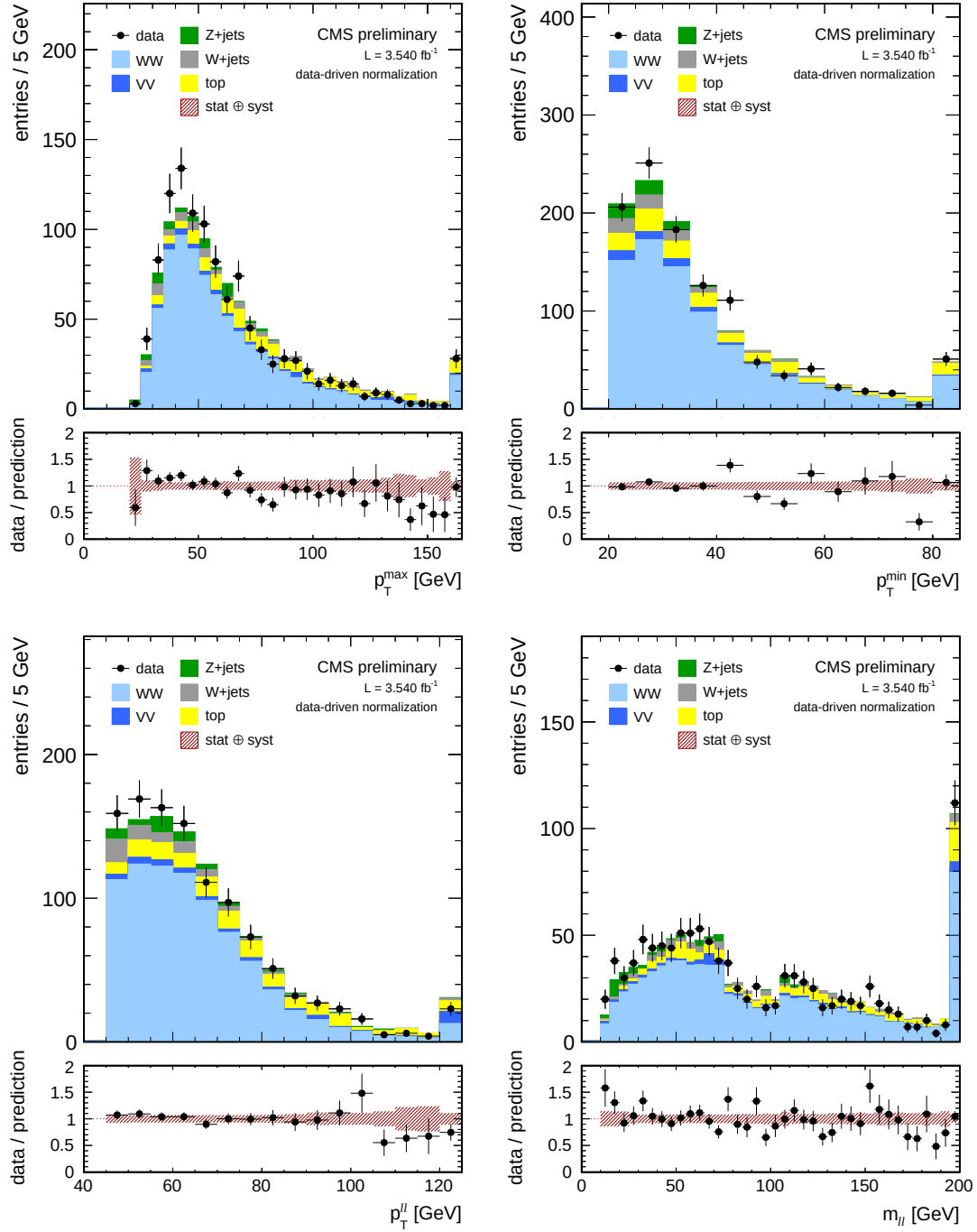


Figure 1: Distributions of the leading lepton transverse momentum ($p_T^{\max}$), the trailing lepton transverse momentum ($p_T^{\min}$), and the dilepton transverse momentum ($p_T^{\ell\ell}$) and invariant mass ($m_{\ell\ell}$) at the final selection level. The W^+W^- contribution is scaled to the CMS measurement presented in this physics analysis summary. All four channels (ee , $\mu\mu$, $e\mu$ and μe) are combined, and the uncertainty band corresponds to the statistical and systematic uncertainties on the predicted yield.

of variations in the parton distribution functions and the value of α_s on the predicted cross sections were derived by following the same prescription as for the signal acceptance. Including

the experimental uncertainties gives a systematic uncertainty of around 10%. The total uncertainty on the background estimations is about 13%, which is dominated by the systematic uncertainties on the normalisation of the top quark and $W + \text{jets}$ backgrounds. The luminosity measurement uncertainty is 4.4% [27].

A summary of the uncertainties taken into account in this analysis is given in Table 2.

Table 2: Relative systematic uncertainties on the estimated signal and background yields, in percent. Data-driven background uncertainties are separated in statistical component of the control sample and systematic component of the extrapolation, (stat.) $\oplus$ (syst.).

	$q\bar{q} \rightarrow WW$	$g\bar{g} \rightarrow WW$	top	W+jets	WZ+ZZ	$Z/\gamma^* \rightarrow \ell\ell$	W+ γ	W+ γ^*
luminosity	4.4	4.4	-	-	4.4	-	4.4	-
trigger efficiency	1.5	1.5	-	-	1.5	-	1.5	-
lepton id efficiency	2.0	2.0	-	-	2.0	-	2.0	-
muon momentum scale	1.5	1.5	-	-	1.5	-	1.5	-
electron energy scale	2.5	2.5	-	-	1.9	-	2.0	-
E_T^{miss} resolution	2.0	2.0	-	-	2.0	-	2.0	-
jet veto efficiency	4.7	4.7	-	-	4.7	-	4.7	-
pile-up	2.3	2.3	-	-	2.3	-	2.3	-
top normalisation	-	-	$10 \oplus 15$	-	-	-	-	-
W+jets normalisation	-	-	-	$19 \oplus 36$	-	-	-	-
Z normalisation	-	-	-	-	-	$15 \oplus 24$	-	-
W+ γ normalisation	-	-	-	-	-	-	30	-
W+ γ^* normalisation	-	-	-	-	-	-	-	30
PDFs	2.3	0.8	-	-	5.9	-	-	-
higher order corrections	1.5	30.0	-	-	3.3	-	-	-
sample statistics	1.1	3.1	-	-	4.1	-	8.4	8.4

6 Results

Counting the number of events in the signal region, the W^+W^- yield is calculated by subtracting the estimated contributions of the various SM background processes. The signal efficiency times acceptance averaged over all lepton flavors including τ s is found to be $(3.22 \pm 0.22 \text{ (total)})\%$.

The total background yield is $275.2 \pm 14.9 \text{ (stat.)} \pm 31.2 \text{ (syst.)}$ events and the total number of events observed is 1111. Using the $W \rightarrow \ell\nu$ branching ratio of (0.1080 ± 0.0009) from Ref. [28], the W^+W^- production cross section in pp collision data at $\sqrt{s} = 8 \text{ TeV}$, is calculated to be

$$\sigma_{WW} = 69.9 \pm 2.8 \text{ (stat.)} \pm 5.6 \text{ (syst.)} \pm 3.1 \text{ (lumi.) pb.}$$

The statistical uncertainty is due to the total number of observed events. The systematic uncertainty includes both the statistical and systematic uncertainties on the background prediction, as well as the uncertainty on the signal efficiency. This measurement is consistent with the SM expectation of $57.3 \left({}^{+2.4}_{-1.6} \right) \text{ pb}$ [1]. The difference between the measured and the theoretical value is $12.6 \pm 7.3 \text{ pb}$, equivalent to $(22 \pm 13)\%$ of the theoretical value. (Experimental and theoretical uncertainties have been added in quadrature.)

Finally, the measurement of the WW production cross section is interpreted in terms of the ratio with the Z production cross section. The Z process is measured using events passing the same lepton selection as the WW measurement, that fall within the Z mass window in the e^+e^- and $\mu^+\mu^-$ final states. The theoretical expectation of the ratio σ_{WW}/σ_Z at 8 TeV is

$(1.72 \pm 0.08 \text{ (scale)} \pm 0.01 \text{ (PDF)}) \times 10^{-3}$, where the Z cross section at NNLO is calculated using FEWZ [29]. The efficiency and the acceptance of the Z selection are obtained using the MADGRAPH MC sample, and the extrapolation factor to the $60 \text{ GeV} < m_{\ell\ell} < 120 \text{ GeV}$ mass range is obtained using FEWZ. The ratio of the W^+W^- efficiency times acceptance to the Z efficiency times acceptance is found to be $0.112 \pm 0.010 \text{ (syst.)}$. The systematic uncertainty on the efficiency ratio takes into account the correlation between those uncertainties that are common to both terms. By using the branching fraction for $Z \rightarrow ee/\mu\mu$ of $(6.73 \pm 0.008)\%$ from Ref. [28], the WW to Z cross section ratio is found to be,

$$\frac{\sigma(WW)}{\sigma(Z)} = (2.09 \pm 0.18 \text{ (syst.)} \pm 0.09 \text{ (stat.)}) \times 10^{-3}. \quad (1)$$

The systematic uncertainty includes the theoretical and experimental systematic uncertainties on the ratio of the efficiency times acceptance for the two processes, as well as the total uncertainty due to the subtraction of background processes. The statistical uncertainty is due to the total number of observed events. The difference between the measured value of σ_{WW}/σ_Z and the theoretical one is $(22 \pm 13)\%$, neglecting correlations between theoretical uncertainties that contribute to both values.

7 Conclusions

This paper reports a measurement of the W^+W^- cross section in pp collisions at $\sqrt{s} = 8 \text{ TeV}$. The measurement is performed in the $W^+W^- \rightarrow \ell\nu\ell\nu$ decay channel, based on a total integrated luminosity of 3.54 fb^{-1} . The measured cross section is $69.9 \pm 2.8 \text{ (stat.)} \pm 5.6 \text{ (syst.)} \pm 3.1 \text{ (lumi.) pb}$, to be compared with the Standard Model NLO prediction of $57.3 \left({}^{+2.4}_{-1.6} \right) \text{ pb}$ [1].

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